DISTROBOX to login Linux with uniform ROSS code

↳ to pull the "image" and have all installation and libraries of ROS → same environment...

POLITECNICO MILANO 1863

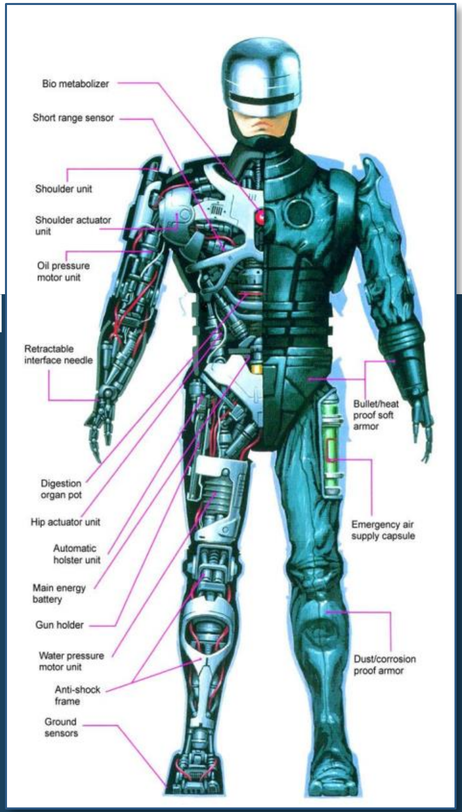
how to measure position/velocity

(Perception / Mapping / Localization) => SENSORS importance

+ overview of ACTUATOR used for motion like electrical motors

Robotics

Sensors & Actuators



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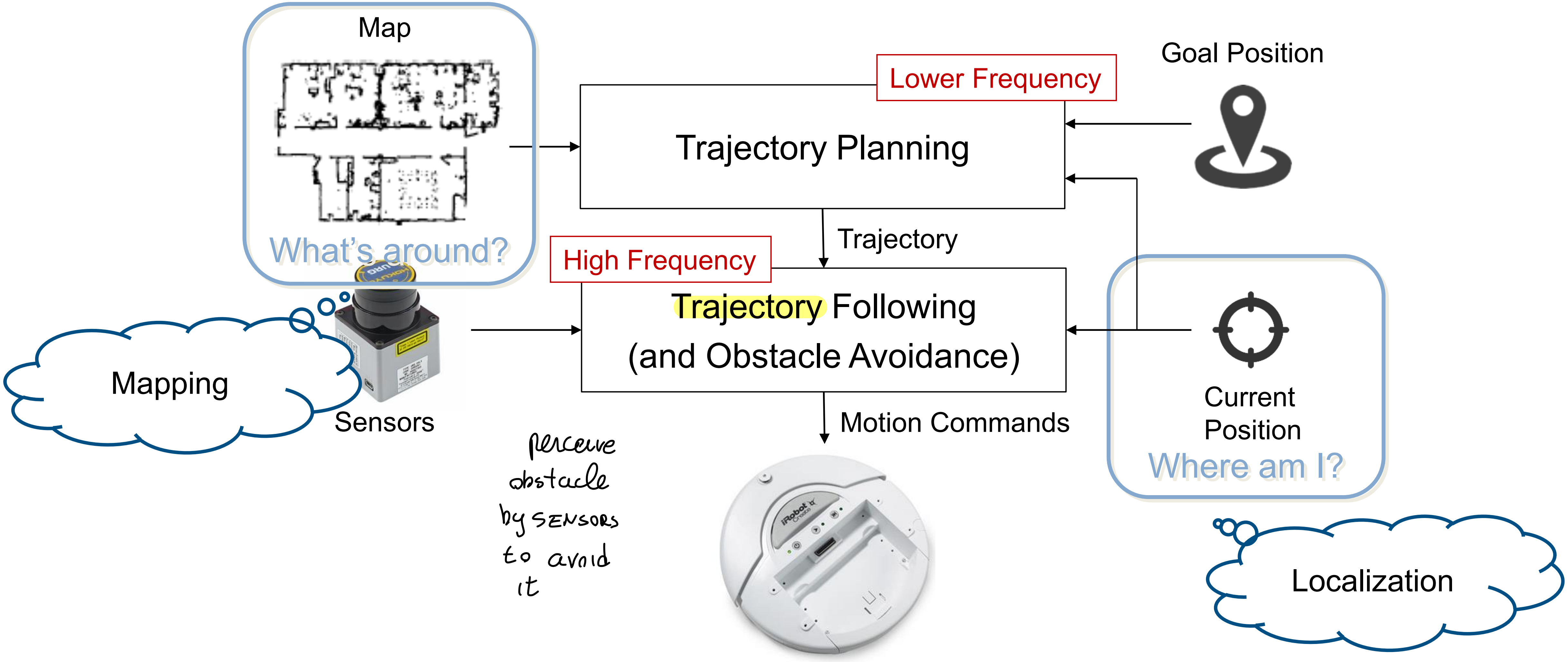
A Simplified Sense-Plan-Act Architecture (Reference architecture)



perceive obstacle by sensors to avoid it to avoid collision → know ROBOT POSITION,

$\begin{cases} \text{traj update @ high freq} \simeq 100\text{Hz} \\ \text{position update @ low freq} \simeq 0.5\text{Hz} \end{cases}$

- how to compute ROBOT POSITION → LOCALIZATION

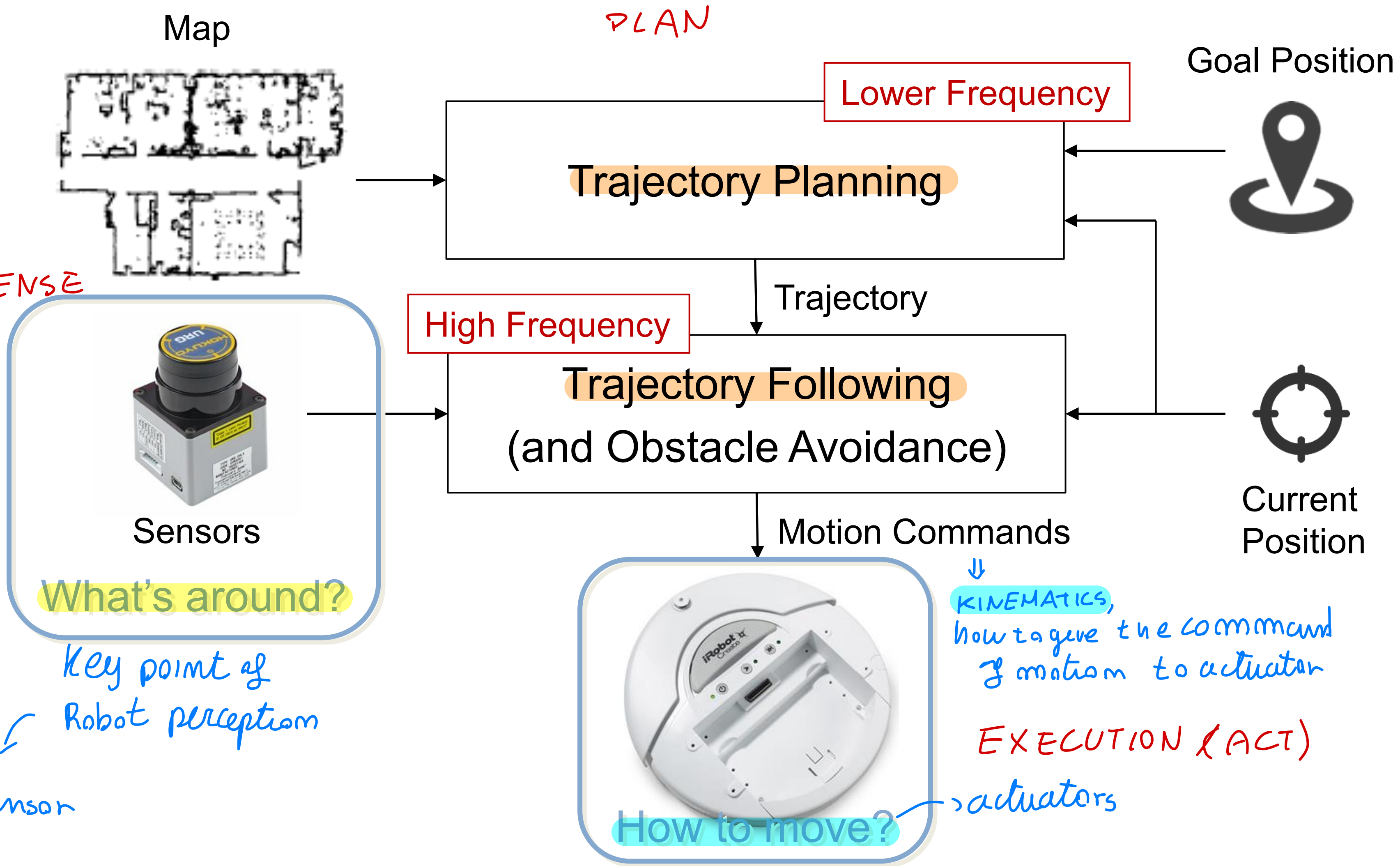
- how to know map around → MAPPING

how it works + implementation in ROS

Sensor model: probability of getting a certain measurement on a certain condition

?

A Simplified Sense-Plan-Act Architecture



PLAN

SENSE

key point of Robot perception

different model of the sensor

KINEMATICS, how to give the command of motion to actuator

EXECUTION (ACT)

actuators

What does it make an autonomous robot?

(ALGORITHMS to decide the better path)

(Sensor) ← Sense - Act - plan
ARCHITECTURE

↳ autonomy by cycle of sense/plan/act

Sensors perceive:

(EXAMPLE)

- Internal state of the robot (proprioceptive sensors)
  - → • battery state of charge (the ROBOT has power?)
  - • Robot speed / position respect ref plane (INTERNAL Respect external Ref)
  - • accelerometer look on F on ROBOT
- External state of the environment (exterocemptive sensors) → used to perceive of obstacle in front (LIDAR)/(RADAR)/(CAMERAS)
  - object AROUND ROBOT ⇒ environment

(part which has an effect on environment)

Effectors modify the environment state

- Match the robot task (e.g., wheels, legs, grippers)

(the GRIP of ROBOT affect env. related to the task) → end effector GRIPPER
↳ specificity of the TASK

Actuators enable effectors to act (device to actuate)

- E.g., passive actuation, motors of various types

(motion between points)

device which make the effectors do the Job →
- • electrical motor
- • stepper motor
- ecc...
- • pneumatic actuation

internal → control Robot action
external → detect characteristic external usefull for Robot

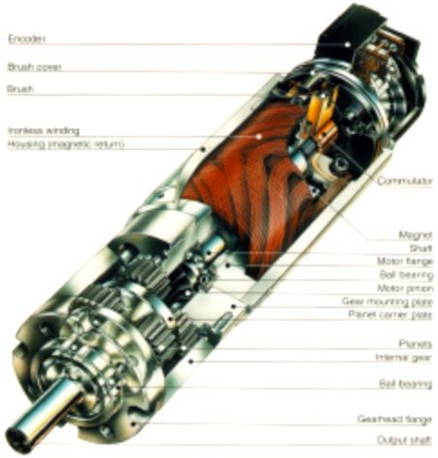
Type of Actuators

Actuators enable effectors to act

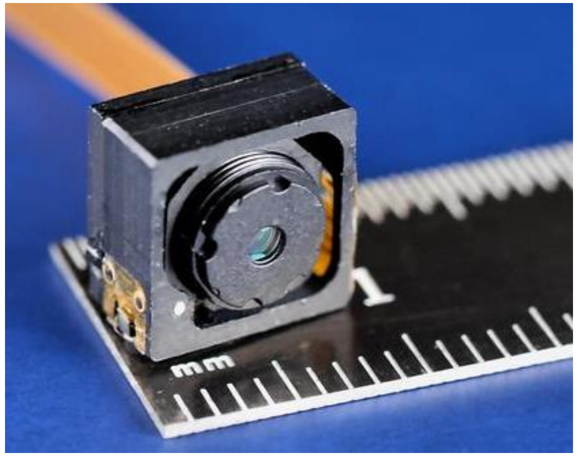
most used, even on car

- Electric motors
- Hydraulics
- Pneumatics
- Photo-reactive materials
- Chemically reactive materials
- Thermally reactive materials
- Piezoelectric materials

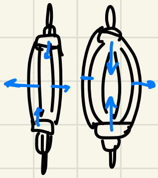
↳ adjust camera focus for ex



Piezoelectric



- **elect motors**: elect power converted into torque
- **Hydraulic**: (motor / trasmission) → used in some application
  (thermal engine) → the device performing the action is hydraulic
  controlled by modulating the pressure
- **Pneumatic** controlling air (fluid) like artificual muscle
  which shrink



work like muscle... enlarge / shrink ecc..
Emulate artificial muscle
(humanoid Robot) only used to pull (shrink)
NOT able to push!

+ others using other phisical quantities

(shape memory material, comeback to original state by heating)

↓

Used to research, to obtain more complex Robotic systems

Most Popular Actuators

used a lot by (BOSTON DYNAMICS)

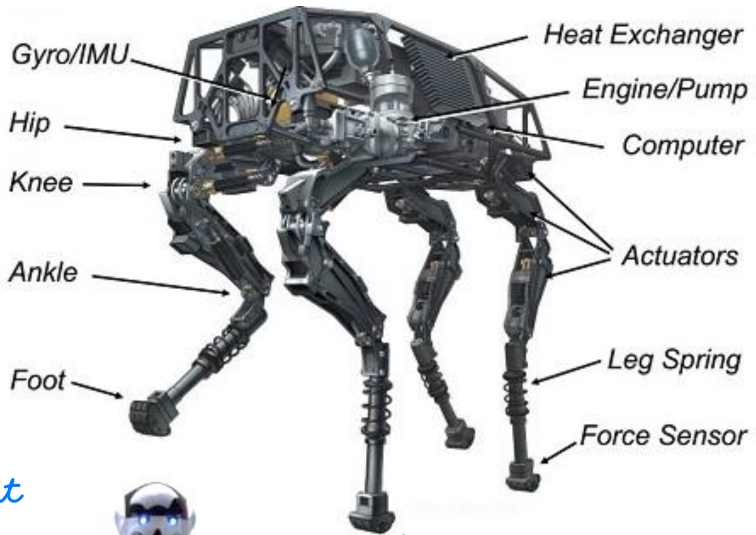
First robots used hydraulic and pneumatic actuators

- Hydraulic actuators are expensive, weighing, and hard to maintain (big robots) → had to be able to absorb shoks + very fast
  high load
- Pneumatic actuators are used for stop-to-stop applications such as pick-and-place (fast actuation)

long extension in small time (FAST Actuators)

↳ not used INDUSTRIALLY for motion point to point

leg with hydraulic act. + thermal engine "Big dog"

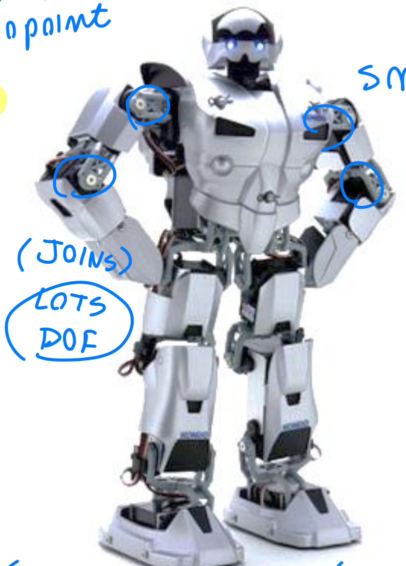


Nowadays most common actuators are electrical motors

- Each joint has usually its own motor (and controller)
- High speed motors are reduced by (elastic) gearing
- They need internal sensors to be controlled
- Stepper motors do not need internal sensors, but when an error occurs their position is unknown

Gear to transmit

small ROBOTS



(JOINS) LOTS DOF

(1 elect motor ∀ JOIN)

elect mot: we contral position /speed
not simply by ON/OFF.. → but with internal contral

ther are also stepper motor which don't need internal contral
(like stepper printer)

ELECTRICAL MOTORS

DC Motors

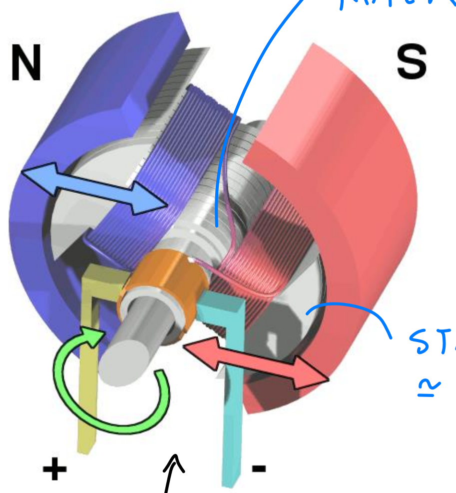
easy to CONTROL + BUILD
cheap ...

- Direct Current (DC) motors (most used)
  - Convert electrical energy into mechanical energy
  - Small, cheap, reasonably efficient, easy to use

How do they work? simple architecture

- Electrical current traverses loops of wires mounted on a rotating shaft
- Loops of wire generate a magnetic field which reacts against the magnetic fields of permanent magnets placed around
- These two magnetic fields push against one another and the armature turns

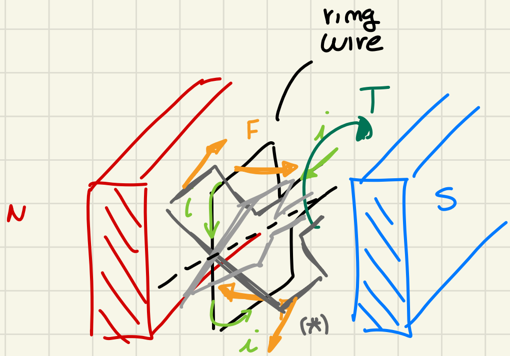
ROTOR



STATOR
≃ permanent magnet usually

BRUSHES ⇒ contact to switch between wires!

DC motor ⇒ STATIC PART (N/S) permanent magnet



magnetic field inside

magn field pushed by N attracted by S and viceversa (F)

↓

{ torque guenerated by the wire T }

using multiple wires (*)

↓ to reduce ripple and have a smoother Torque

{ Keep ROTATING around ROTATION axis }

most of elect motors are based on this principle

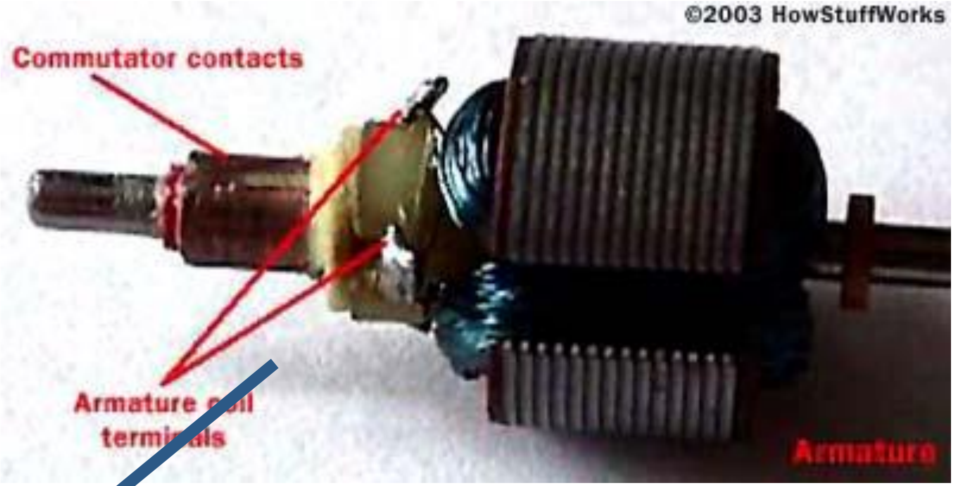
→ depending also on How the switch happen

- Sometimes the Rotating part can stay outside

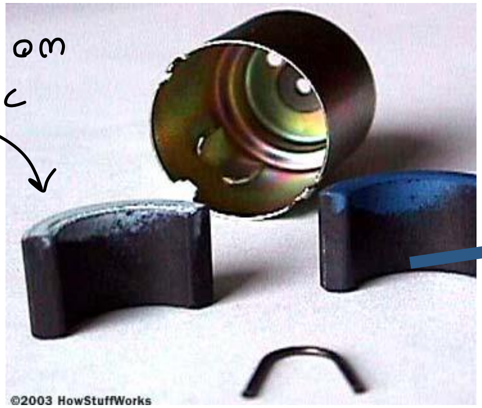
DC Motors

higher charge → higher Torque ↓ higher speed

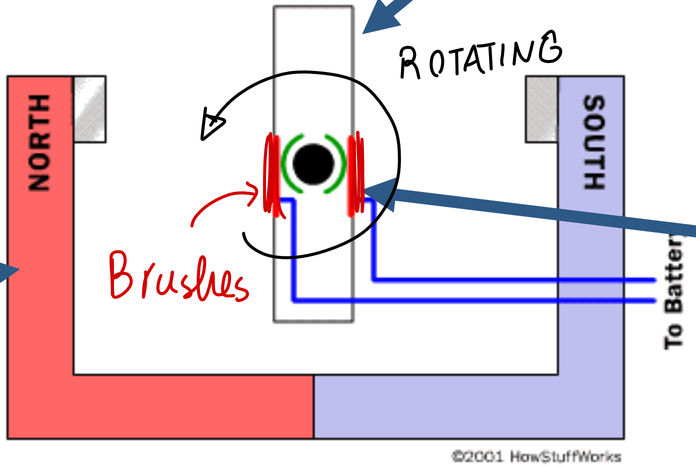
outside static part

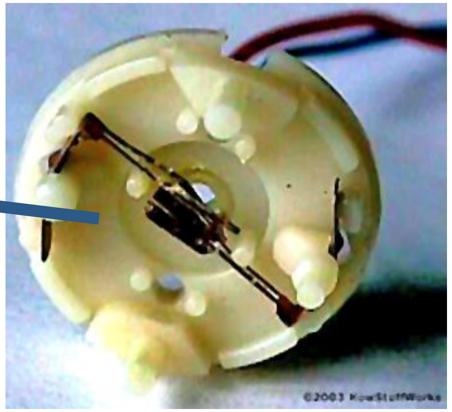


magnets on the STATIC part of elect motor



ROTATING





usually this DC motor ROTATE very fast

3KRPM usually → T limited but high Ω_m

usually GEARINGS (trasmission)

so $\Omega_m \downarrow \Rightarrow T \uparrow$

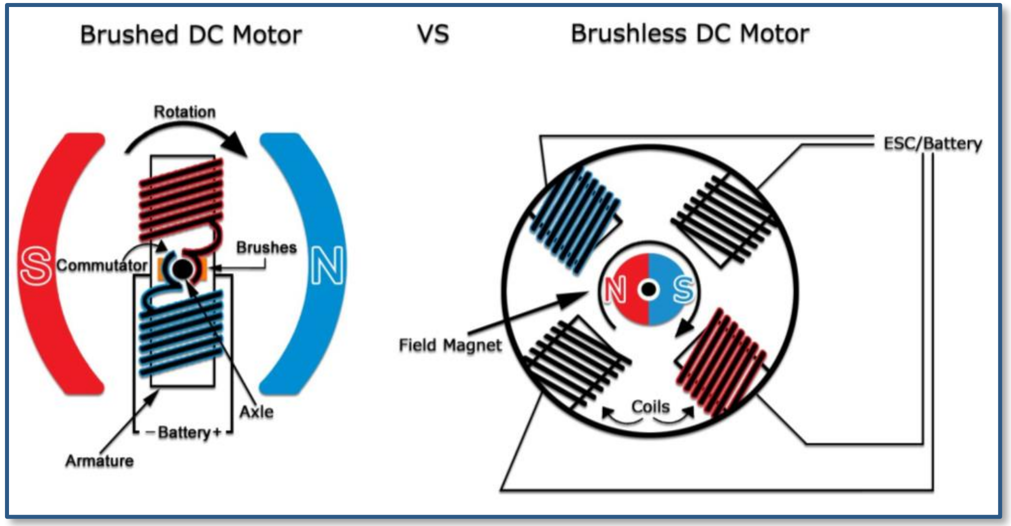
DC Motors: Brushed and Brushless Motors

usually wires (electromagnet) on stator instead of permanent one → when using brushes: continuous contact!

to have feedback loop you need sensors to measure Ωm... speed/position sensors used on ROBOTICS

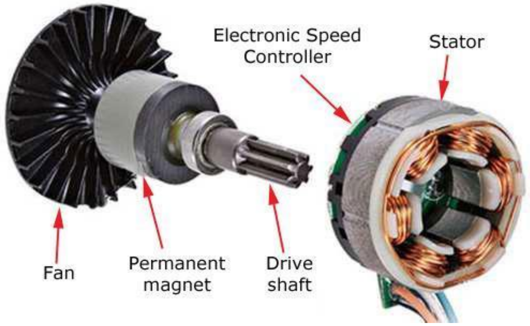
1. Brushes used to change magnetic polarity, they're cheap but ..
   issues ↓ (used to switch)
   - Brushes eventually wear out → when long life time, the brushes can WEAR too much
   - Brushes make noise
   - Limit the maximum speed
   - Hard to cool → heat generated by contact
   - Limit the number of poles

⇓ to solve Brush issues



2. Brushless motors overcome these problems but they are more expensive
   - Brushes are replaced by computer
   - Permanent magnets on the rotor
   - Electromagnets on the stator

↑ to switch between wire you can use a computer control



BRUSHLESS → let magnetic part Rotate while wire static
but the Force pull/ push the ROTOR that follow
the STATOR
outside stator rotate, followed by
rotor inside...

but starting to
ROTATE ⟹ brushless mechanicsm

magnet Rotating and use a computer controller to
swith between wires to let it ROTATE

(Refersed function)

3 **Stepper motors**

in this case we DON'T need sensors... on small moovement
you don't need a sensor here
⇓ like brushless motor working..
electromagnet switched to generate one ROTATION STEP

(actuate for each step, degree of Rotation)

shaft motion one step at a time (+ micro moovement) (+ stop motion)

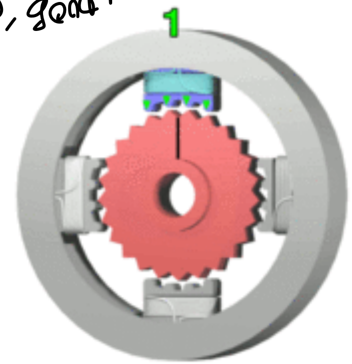
A *stepper motor* is a brushless, synchronous electric motor that converts digital pulses into mechanical shaft rotations.

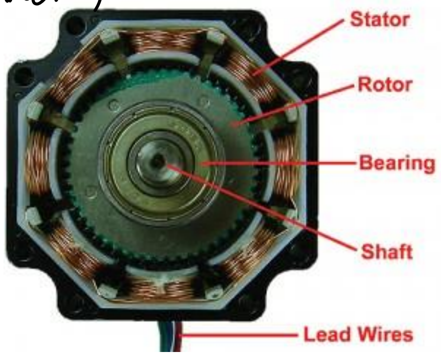
lot used, cheap, good Resolution

- ✅ Rotation angle proportional to input pulse
- ✅ Full torque at standstill (energized windings)
- ✅ Precise positioning and repeatability
- ✅ Response to starting/stopping/reversing
- ✅ Very reliable (no contact brushes)
- ✅ Allow open-loop control (simpler and cheaper)
- ✅ Allow very low speed synchronous rotation with a load directly coupled to the shaft.
- ✅ Wide range of rotational speeds
- ❌ Require a dedicated control circuit → ∀ STEP
- ❌ Use more current than D.C. motors → require more power
- ❌ Torque reduces at higher speeds → Ω limit / No high speed
- ❌ Resonances can occur if not properly controlled.
- ❌ Not easy to operate at extremely high speeds

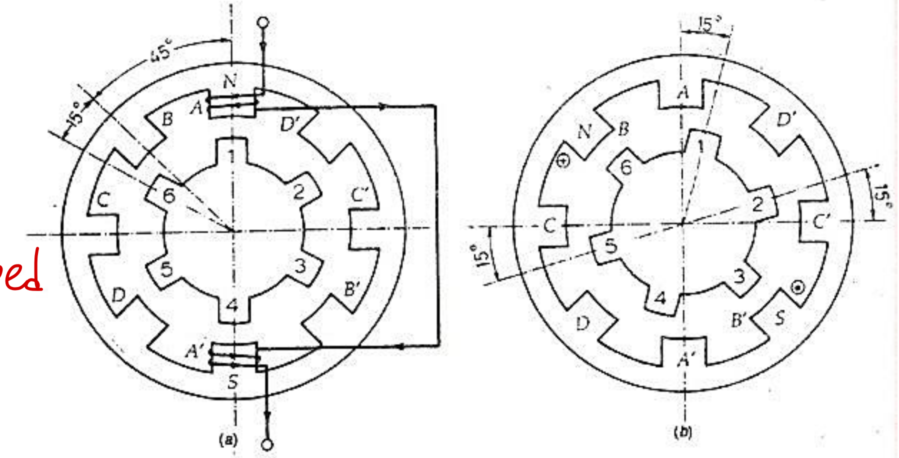
NOT for CONTINUOUS MOTION!

• NOT very accurate.. you can lose steps





$$\alpha = \frac{N_s - N_r}{N_s N_r} \times 360°$$



Servo Motors

used into ROBOTS
very similar to: COMBINATION of MOTOR + SENSOR
(vehicles radiocontrolled) (position control - actuations)

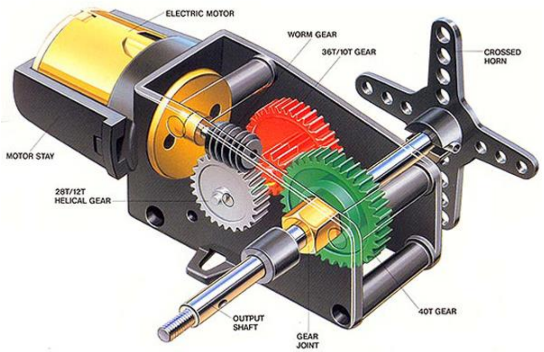
any Joint can be < 180°
for humanoid robot
+ position control of JOINT

motor with precise position but limited speed

"Servo": specialized motors that can move their shaft to a specific position

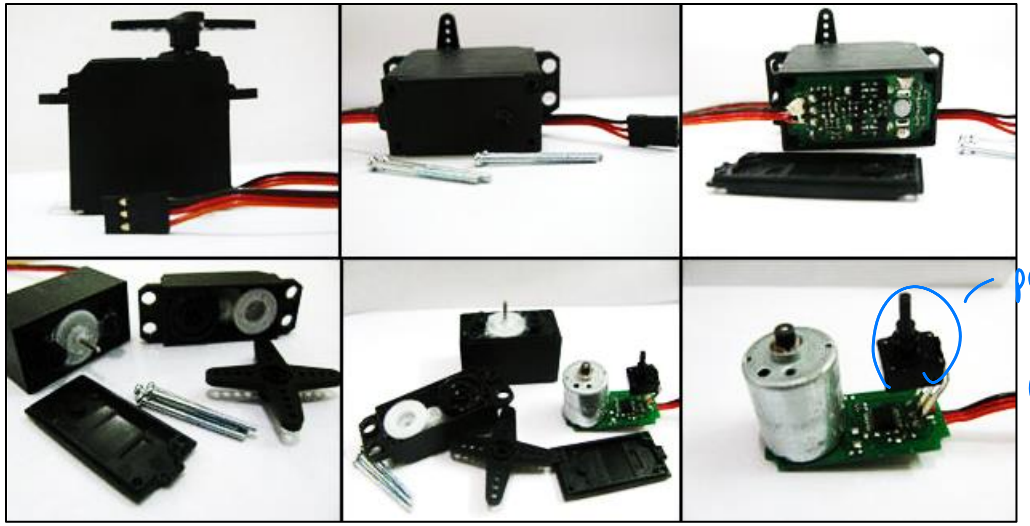
- Used in hobby radio control applications
- Measure their own position and compensate for external loads when corresponding to a control signal

sensor measuring shaft position



Servo motors are built from DC motors by adding

- Gear reduction
- Position sensor
- Control electronics



potentiometer sensor
+ RETROACTION
(feedback loop to position shaft)

Shaft travel is restricted to 180 degrees
but it is enough for most applications

(MOTOR+SENSOR+CONTROLLER)
↳ moves Robot to reach desired position

close a control loop to reach target set-point POSITION

(somes can control velocity (RARE))

(Pulse Wave modulation)

↓ PWM signal to control servo motors

(NOT classic PWM on brushless motors)

servo: has an embedded controller,

PCB inside the system

on servo motor: you can control speed / position with

voltage / current → with nested control loop

Sensors

"Mobile Robots" moove on wheels, by electrical motors
Voltage input to let Rotate elect motors...
→ how fast they Rotate / position
to know where it is mooved, control motion ⇒ ENCODER : how much Rotate

Sensors allow a robot to accomplish complex tasks autonomously

distinctions

Two main categories

related to ROBOT

- Internal sensors (proprioceptive) 1
- External sensors (exteroceptive) 2

related to ENVIRONM

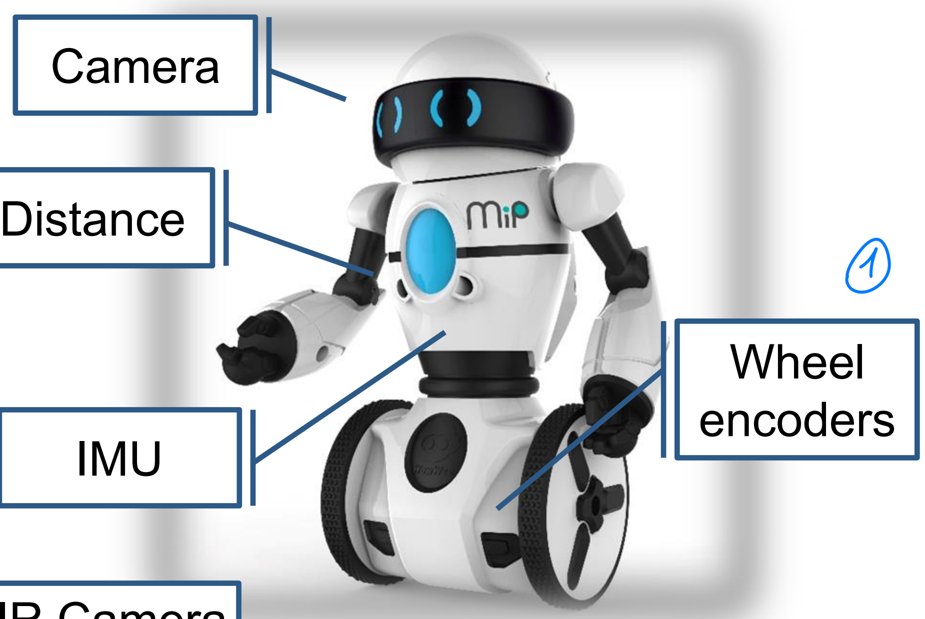
⇓

Other classification

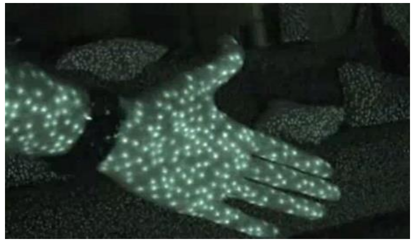
Collect energy

- Passive (measure physical property) vs
  Active sensors (emitter + detector)

↳ inject + receive energy



2 Camera
2 Distance
1 IMU
1 Wheel encoders



it has a camera INFRA RED and RGB (color camera) to collect light

Projector ↓ project pseudo-Random dots... on IR region seen by camera to estimate the scene depth

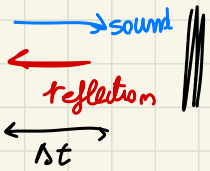
RGB Camera ⇒ PASSIVE sensor, Just sense

Sensor → **Passive** phisical property measurement, JUST measure, like when meas a Temperature I don't need to do things to measure

→ **Active** do something to measure..

(ex) parking sensors on cars beeps when near

⇒ ULTRASOUND sensors



rough estimate of distance (SONAR) by Δt

OR for example with RADAR... OR on Thermal sensor

emission + measurement

usually distance measurements use Active sensors...

Inject energy and reflect it transform into distance

(except for STEREOVISION which collect images and detect distance, PASSIVE)

Encoders

measure position/speed of motors
OPTICAL ENCODERS uses stripes encoded
(counting the stripes they have passed through)

set of holes, with light emitted to measure how many steps you see.. to see the holes in front of light.. you know how much it moves

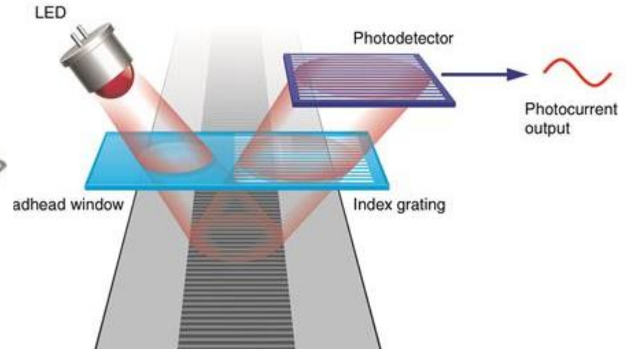
black-white hole.. light go through and we detect it!

An encoder converts motor/joint rotary motion or position into electronic pulses

knowing Resolution you know motion

Linear encoders

- Consist of a long linear read track, together with a compact read head



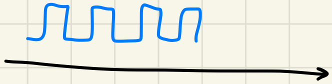
(ROTARY MOOVEM.)

Rotary encoders

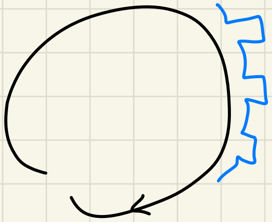
usefull for ROTATING desplacements
when using Gears.. wheel on stripes

- Both for rotary and linear motion (in conjunction with some mechanism) convert rotary motion into electrical signals
- They can be incremental or absolute

attached on shaft MOTOR to measure Rotation

linear 

Rotatory

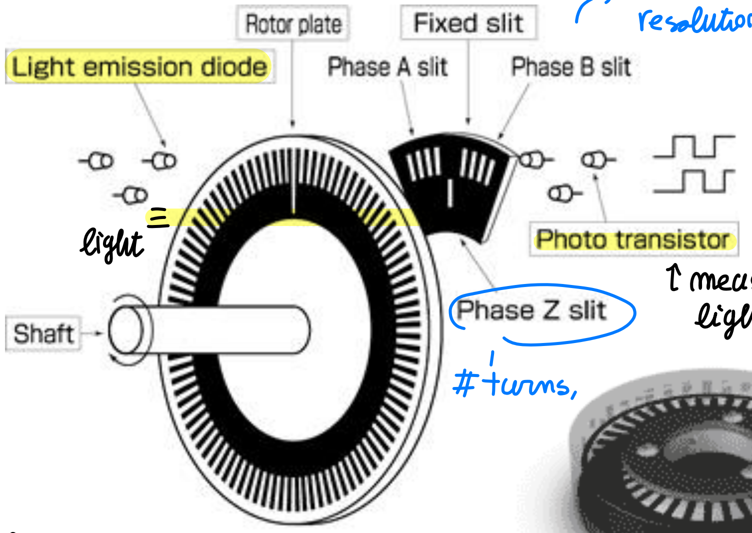


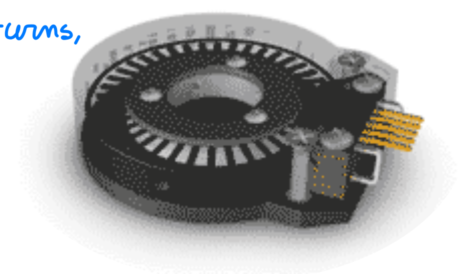
Incremental rotary encoders

square waves generated while moving, obtain the degree of motion by the sensor measurements (1)
and speed depending on frequency of measure square waves
angular speed / position!

It is based on the photoelectric principle

- A disk with two traces (or sensors) where transparent and opaque zones alternate
- The two traces allows to identify rotation direction and increases resolution (quadrature)

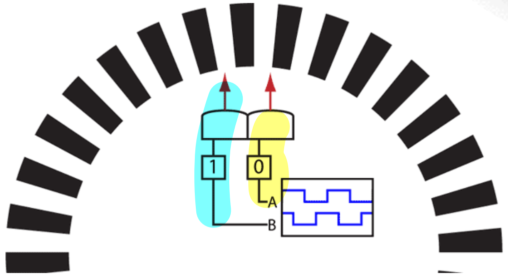


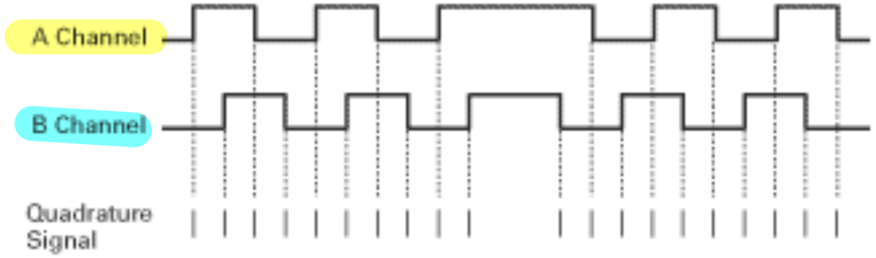


square waves.. how to get information about ROTATION direction

Quadrature technique

- The two signals are shifted by ¼ step
- N, is the number of steps, i.e., the number of light/dark zones, per turn
- Resolution is 360°/4N
  - CCW: 1 1 is followed by 1 0
  - CW: 1 1 is followed by 0 1





2 channels A/B

(PROS)

↓ you get 4 events ∀ slot ⇒ so get 4 time previous resolution

- 4* resolution with 2 channels ↙

  ↳ quadruple resolution (accurate)

- depending on rotation direction, it changes the overall square form

  ↳ measure rotation

- telling Ø angle you can count ⇒ Reference respect Ø position counting thicks from origin...

  ↓

  you have to set up on ZERO

⇐ we use this increm. encoder to measure <u>speed</u>

↳ IF you need also <u>POSITION</u>

↙

ZERO techniques to go @ Ø ORIGIN each time you switch ON/OFF

⇒ solve this with absolute ROTARY encoder

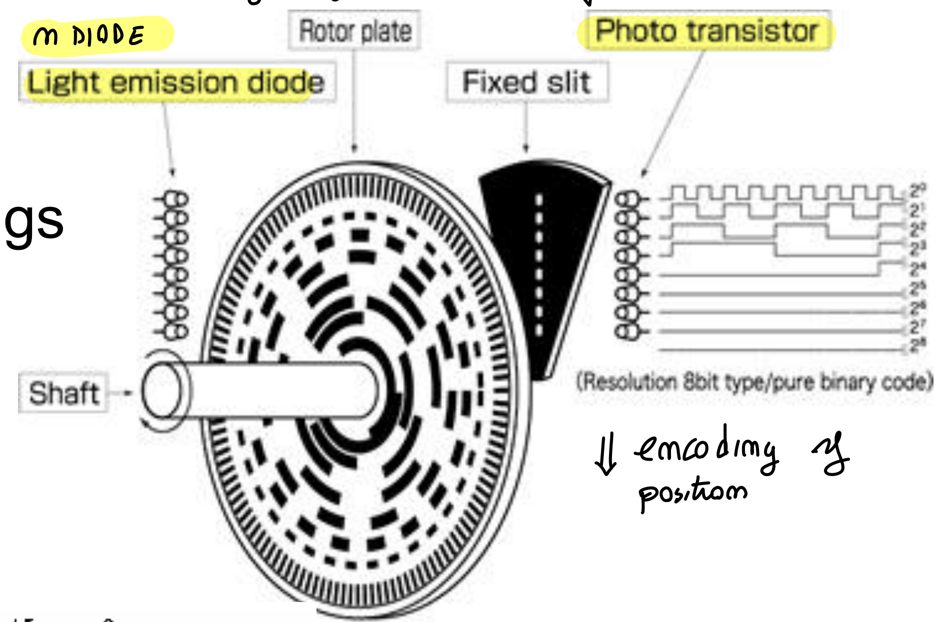
Absolute rotary encoders

same principle of incremental one but more holes and different frequency ∀ circle

each disk ring doubled the previous frequency of bigger disk

n sensors ∀ ring

n DIODE



⇓ encoding of position

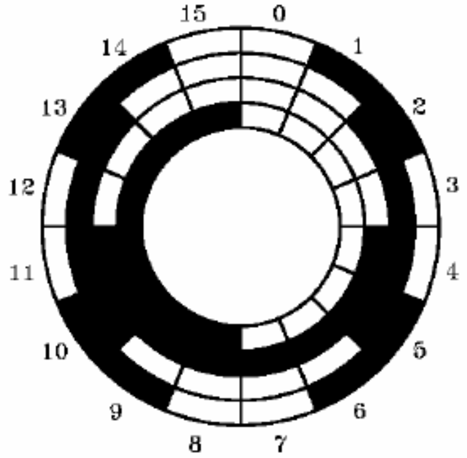
you get in totale 2^n possible angle (Binary)

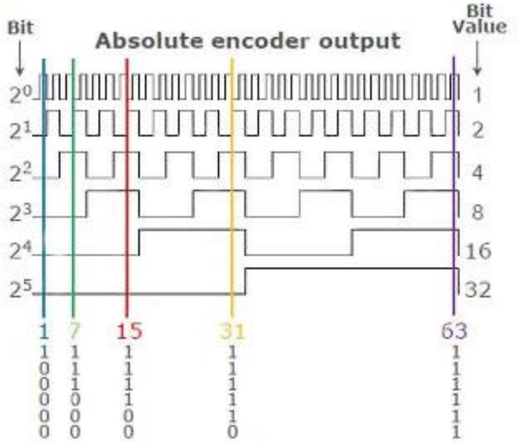
The disk econdes a position

- Transparent and opaque areas on concentric rings
- For an N-bit word there are N rings
- Absolute resolution: $360°/2^N$
- In robotic applications at least 12 rings are used (360°/4096) → ≃ 0,1° resolution
- Binary codes with single variations, i.e., Gray code, are used to avoid abiguities

↳ shuffle order of holes

⇑ QUITE good position, not used in industrial application





Gray coding → so only 1 bit changes between...
so for 1 bit error only 1/10 angle error!

only 1 bit changing ⇒ only 1 error

While of binary error ⇒ IF you make error
on significant BIT of classical DIGITAL CODING
you get HIGH error!

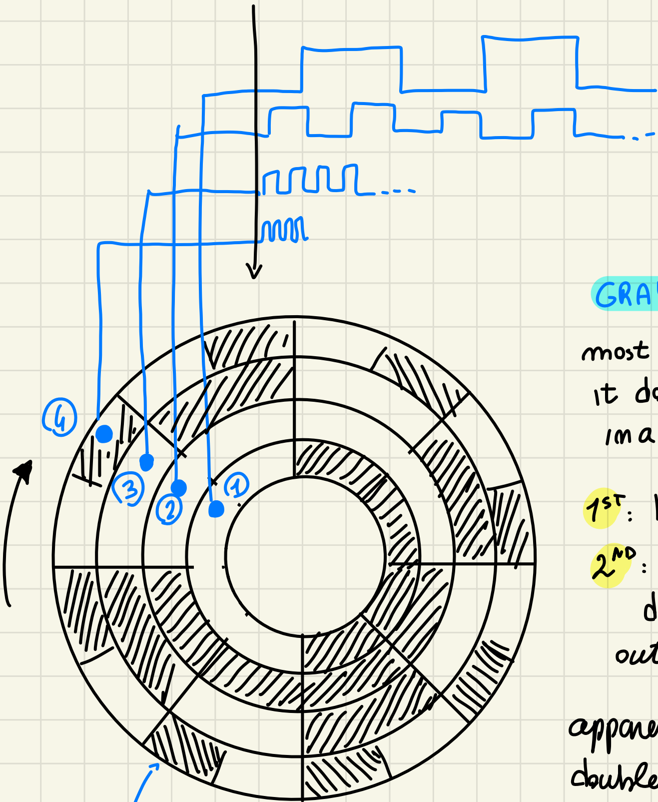
↓

While with Grey coding each number
change only 1 bit ... ⇒ small error ∀ revolution
of error on measure

shifting by some
bits... resolution
change chosen properly

Wheel moving by
Read velocity
↑ easy by ENCODER
↓
estimate of position...

Read vertically the bit to decode it



doubled
doubled
doubled

each sector
reffered to a STRIP
of LED

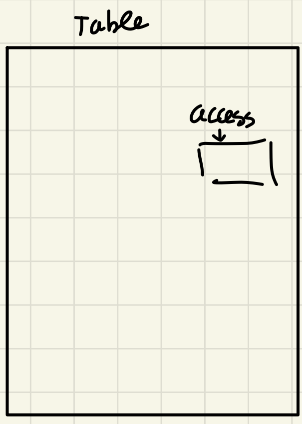
GRAY CODE

most internal 1/2 black, 1/2 white
it double the frequency but
in a complex way

1ST: half black, half white
2ND: 4 pieces, Gray coding
double freq.
out of 4 bit → 2B, 2W

apparently it seems NOT
doubled → in Reality there is
a doubling of bits

↳ Coding with a table of access ⇒



memory
access to
table to
assign the
corresponding
displacement

Distance perception: time-of-flight telemeter

↓ DISTANCE sensors

extrocepting to avoid obstacles
for MAPPING / PERCEPTION

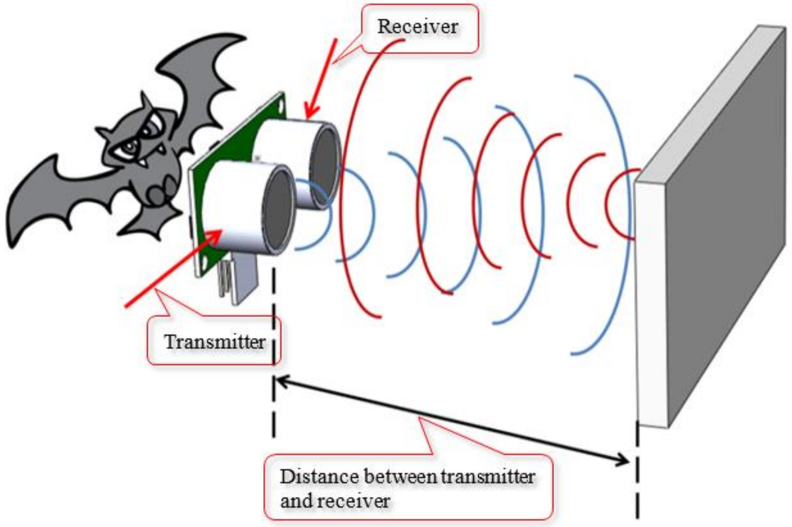
SONAR simpler device...

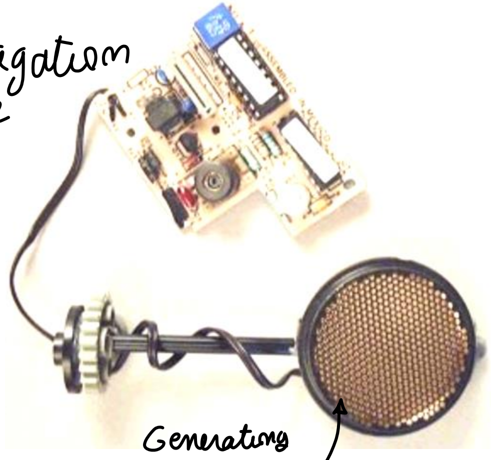
It measures the time between the emitter produces the signal and the detector receives its reflection

- Distance covered by the signal is *2d*
- Time of flight is $\Delta T = 2d/c$

Acoustic waves are used (although light is possible)

- Low speed: v=340 m/s (sound speed)
- Low directionality: 20 – 40° (beam propagation of wave)
- Polaroid ultrasonic sensors (sonar)
  - range 0.3 – 10m
  - accuracy 0.025m
  - cone opening 30°
  - frequency 50 KHz sound frequency





Generating sound

The signal is largely affected by noise with significant reflections …

Sonar: measure the time to go and come back...
so measuring the distance

↓ ACUSTIC

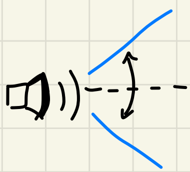
↓ ULTRA SOUND

easy to generate and detect reflected sound

{ACTIVE SENSOR}
↳ you send sound energy to environm.

sound propagate as a cone

come back signal inside that cone...

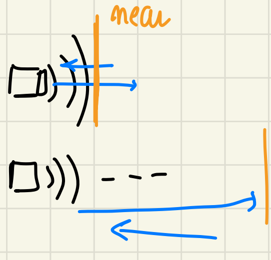


(called)

- "POLAROID sonar" because used on polaroid to adjust the focus depending on people distance

you need fast measure when near...
if FAR obstacle you have to wait

there is a certain frequency of update to re-send the signal



Cheap BUT with lots issues...

ect used in car sensor of parking

(20cm ÷ 2m) range of usage → to see big objects

⇒ Bad quality, Bad resolution

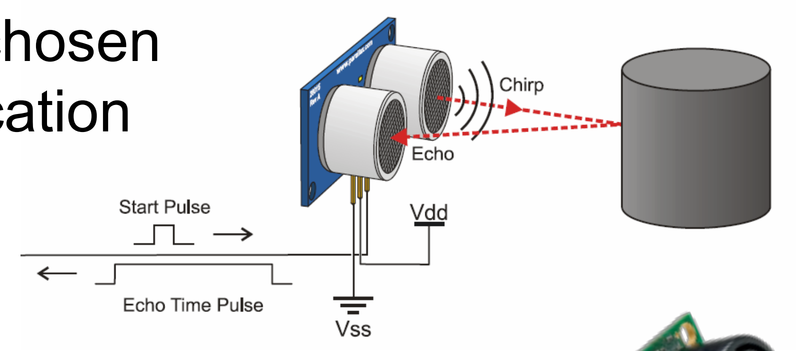
Issues with sonars

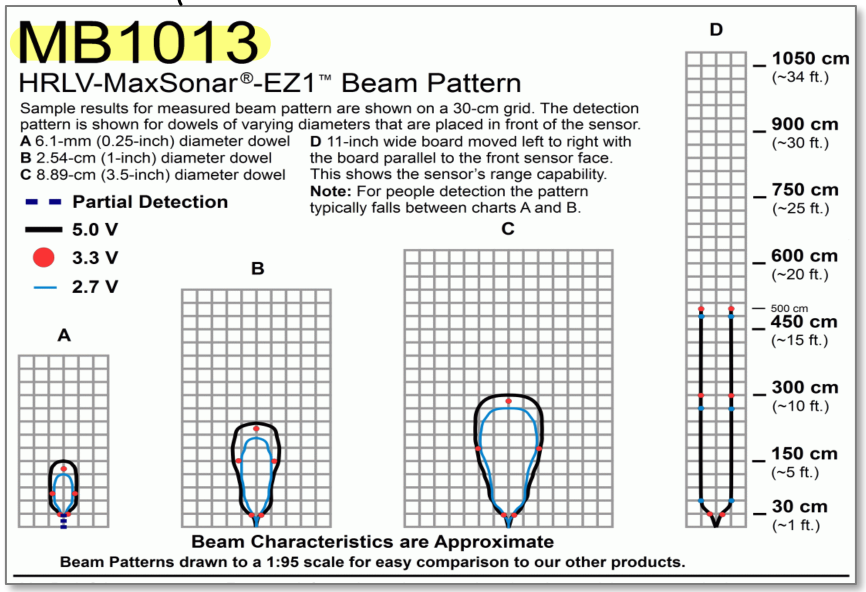
(30° angle is an issue)

ULTRASOUND → has a certain range!

arrive ~30 m with lots Voltage

The range should be chosen according to the application



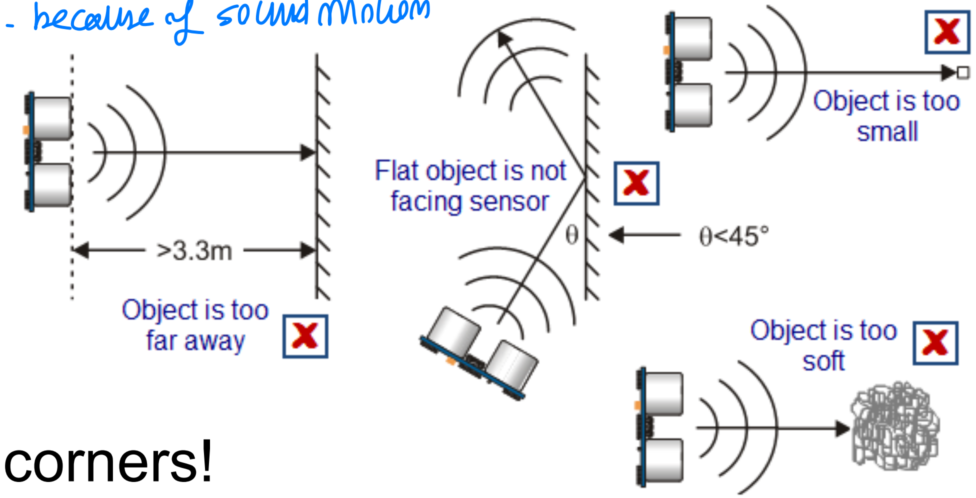


PROBLEMS

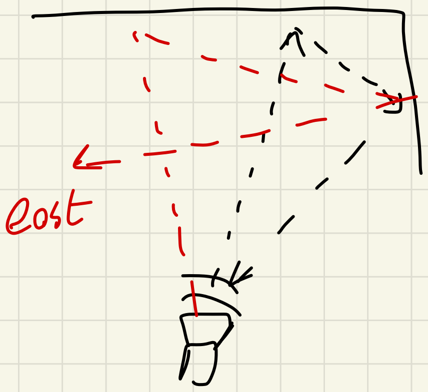
They do not work in all conditions

more readable too often... because of sound motion

- Sampling frequency trade-off
  ↳ between freq and distance
- Reflections against walls
  ↑ sound bounces
- Small objects ↖ hard to detect when reflect less
- Soft objects
  ↳ DON'T Reflect



Rooms may look larger than expected at corners!



longer than
st expected..
bad estimation
OR lose of the signal

sometimes sonars are NOT
enough Good!
NOT for MAPPING...
[depends on final task]!

a mapping with
sonar can be
very bad!

⟹ More
Reliability!

moove away from
Usage of sound... different
Kind of energy
light sensors ⟹

Distance perception: reflective optosensors

conditions where sonars are NOT enough..

INFRARED, Sharp sensor

send ⇄ receive light.. BUT this is high speed, NOT measurable Δt

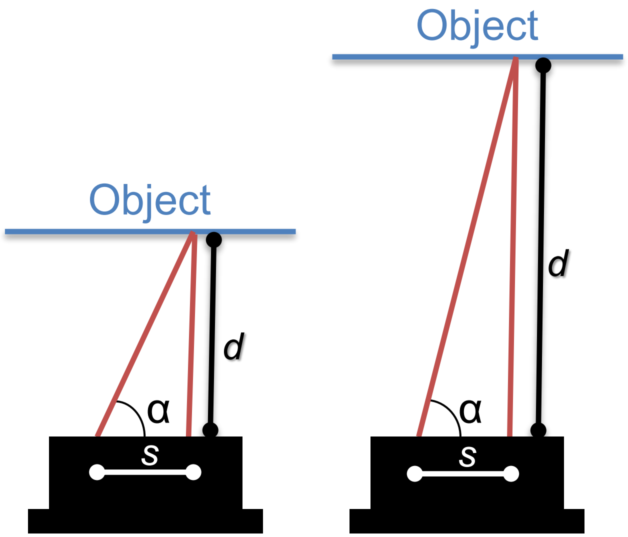
Reflective optosensors are active sensors (e.g., SHARP IR Sensors)

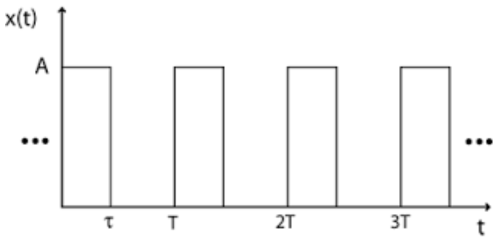
- Emitter: a source of light, e.g., LED (light emitter diode) or IR (infra red)
- Detector: a light detector, e.g., photodiode or phototransistor

It uses triangulation to compute distance

- Emitter casts a beam of light on the surface
- The detector measures the angle corresponding to the maximum intensity of returned light
- Being *s* the distance between the emitter and the detector

$$d = s \cdot \tan \alpha$$





send laser and detect @ which angle the beam is reflected TRIANGULATION!

↓

measure the $\tan(\alpha)$ distance ratio and reconstruct distance!

based on d, s parameter and α angle

⇓

sensors working on IR domain

Infra Red: • silicon excited... so with CMOS easy to be detected

• eyes not excited by its → people don't see it

⇓

IR received by sensors

Sun is an IR source always present, so you try to detect pulses

⇓

pulse encoding

• very cheap, same ultrasound

Don't work:
• mirrors
• glass door
• black obj

} reflective surface problematic!

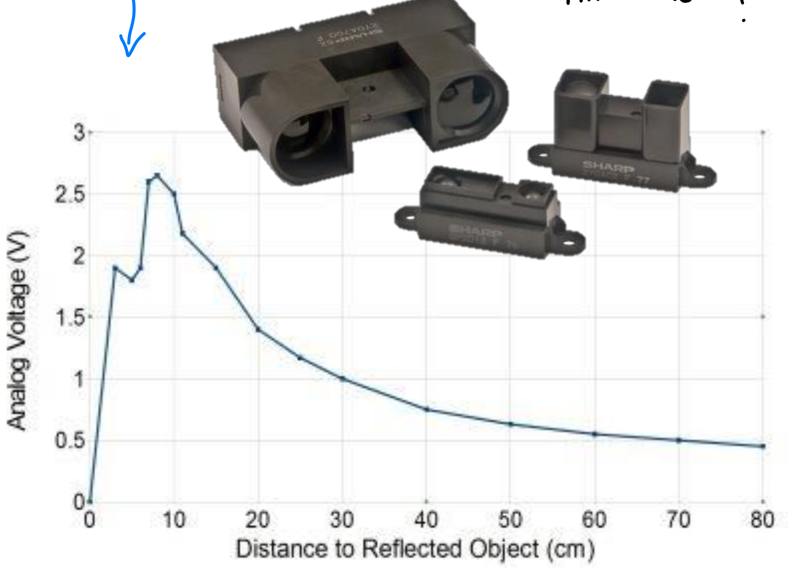
Issues with reflective optosensors

angle issue ... short range ambiguity!

to be sure of close object, where this require to ... usually mounted inside Robots to increase distance all, so it measure more dist!

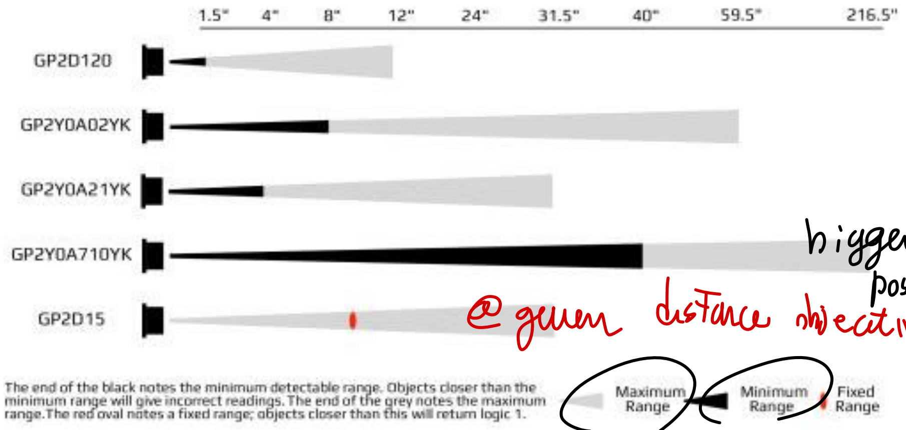
Infrared sensors are relatively cheap and robust but ...

- Non linear characteristics which need to be calibrated
- Have an ambiguity for short range (should be placed in the robot)
- Have fixed ranges / opening angles (requires proper selection)
- May suffer reflections ... sometimes



you can by this sensors dealing with different ranges

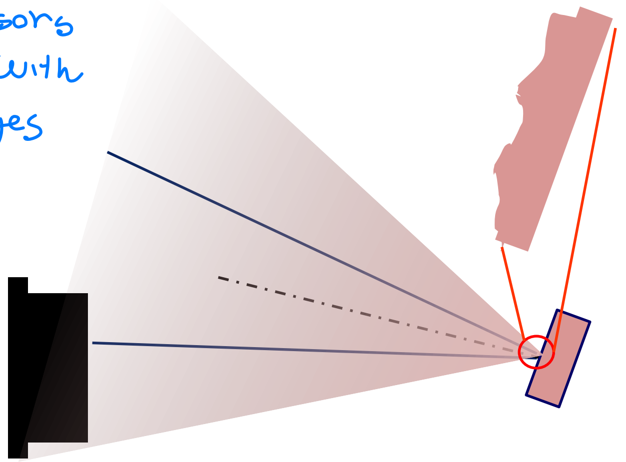
after some 50 cm I cannot measure it



bigger distance possible

@ given distance objective

of the sonar



when close object, narrow spot

REFLECTION! how light come back...
unless mirrors some light
always come back

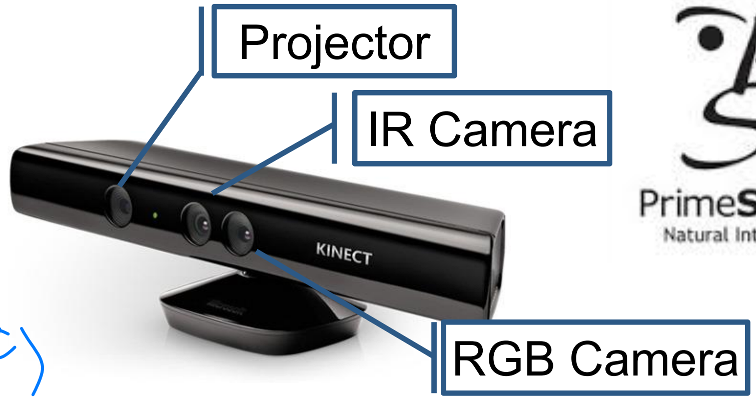
basic sensors...

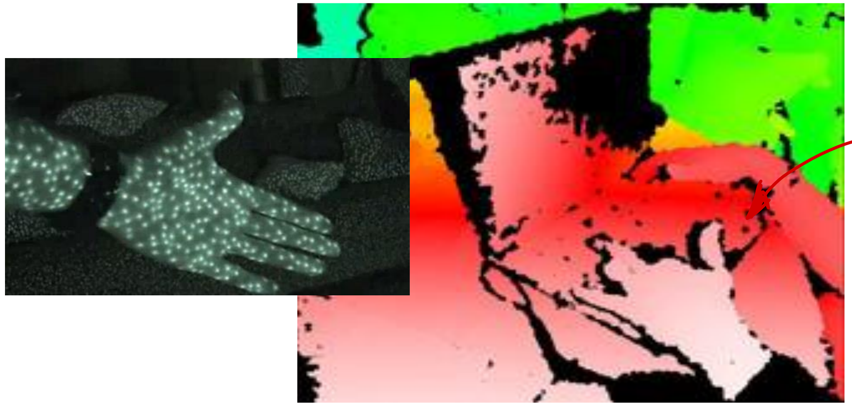
Distance Perception: Kinect

device that changes Robotics

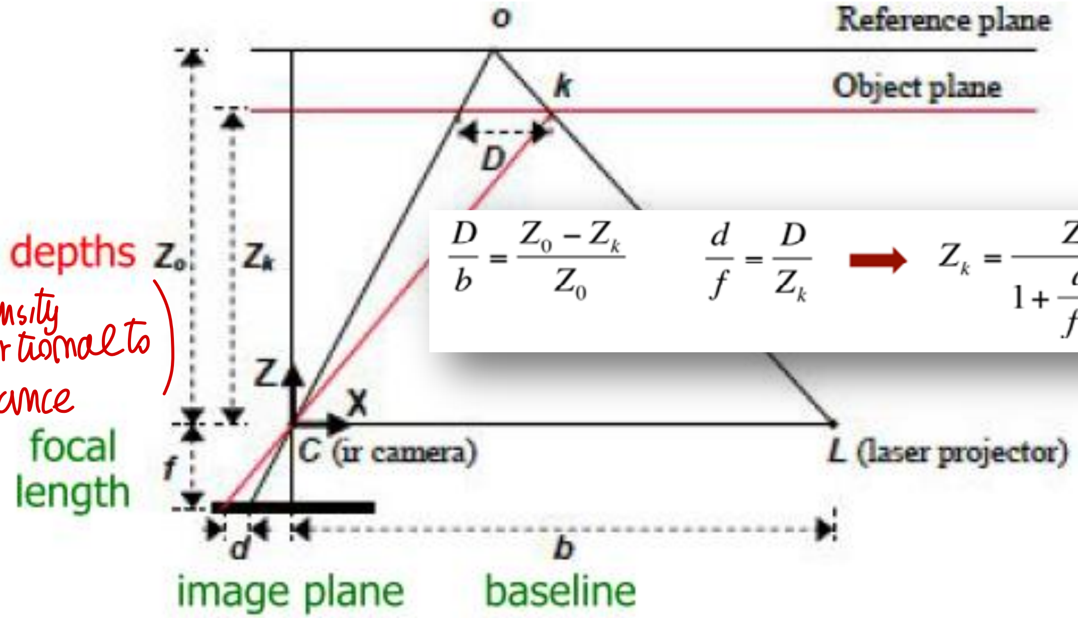
Kinect is a motion sensing input device built by Microsoft (Primesense) for Xbox 360

- 3D scanner
  - Infrared projector
  - Infrared camera (11-bit 640x480)
  - Range 1.2 – 3.5 m (up to 0.7-6 m) (where it works)
  - Angular field of view: 57° h, 43° v
- 30Hz 8-bit RGB camera (640x480)





intensity proportional to distance



$$\frac{D}{b} = \frac{Z_0 - Z_k}{Z_0} \qquad \frac{d}{f} = \frac{D}{Z_k} \quad \Rightarrow \quad Z_k = \frac{Z_0}{1 + \frac{d}{fb} Z_0}$$

IR camera to measure 3D distance of object...
↓
send set / cloud of points and read this → many IR
measured by multiple points
↓
pattern ... estimate

based on TRIANGULATION of many points

pseudo-Random points cloud with unique shape found by camera

{ formula based on triangulation,
Respect a ref plane where collecting points.. }

LOT used on ROBOTICS sensors!

changed robotics
which had to uses stereo
camera ecc...
you project.. stereo
with decoded image to see

Kinect : one of the
most used
sensor! RGB camera also!

(several producers of it like INTEL...)

Some problems outdoors due to sun IR, very noisy
but same problems on mirrors, Glass ecc...

Distance perception: time-of-flight camera

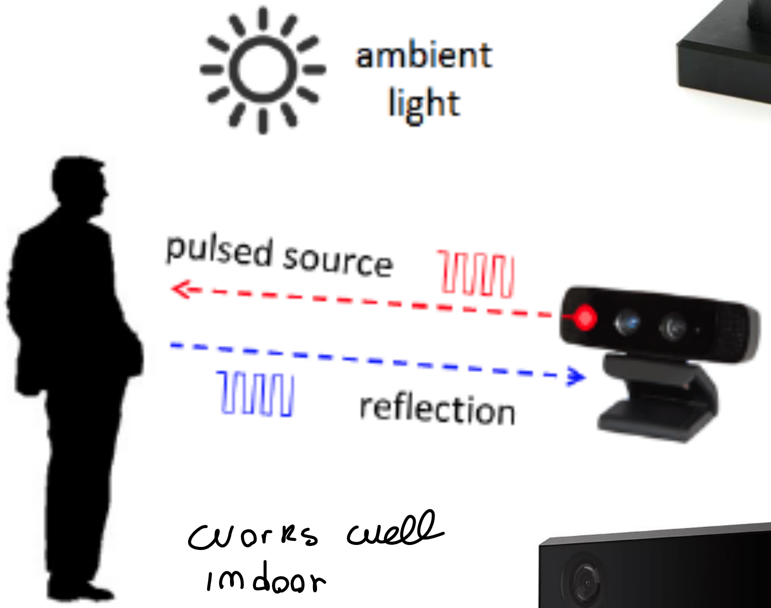
even with Prime sense CLOSE ... respect TRIANGULATION of kinect like
light sensor ⇒ other distance techniq. USE light as SONAR

3D time-of-flight (TOF) cameras

- Illuminate the scene with a modulated light source and observe reflected light
- Phase shift between illumination and reflection is translated to distance

light sources of IR modulated

SWITH RANGER



works well indoor

Microsoft kinect

Some issues exist with these sensors

- Illumination is from a solid-state laser or a near-infrared (~850nm) LED
- An imaging sensor receives the light and converts the photonic energy to electrical current
- Distance information is embedded in the reflected component. Therefore, high ambient component reduces the signal to noise ratio (SNR).

send light beam and measure Δt required

↳ count time required to come back!

BUT high light speed! You modulate light

so that shift of returned light is proportional to dist.

you measure phase shift of In/out

measure reflection → accurate travel time by phase shift!

usually you use LIDAR, SNR or others

Lot of energy on wide spectrum

filter in front of camera to measure energy

most advanced Ranging device..

OUTDOOR not very accurate! because of

noise environment → use light source more powerful

[single camera + AI (deep learn)] geometrically work by experience to build a distance mapping

train the model reconstruct distance

↓

monocular distance estimation

with ML you also need to understand the object type
and guess the distance !
↓
interesting training model
unsupervised !

all seen sensors work indoor

outdoor → energy required high

⇒

2 solutions { • stereo camera, collect 2 images
and triangulation for distance ⟹
• lot light source → LIDAR
sensor

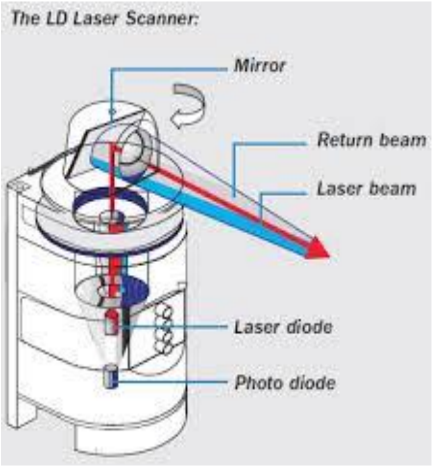
Distance perception: Laser Range Finder

LIDAR (light version of Radar)
LASER RANGE FINDER

work as a single beam time of flight sensor..
measuring Δt to come back → lots energy sent

Lasers are definitely more accurate sensors

- 180 ranges over 180° (up to 360 °)
- 1 to ~~64~~ planes scanned 128
- 10-75 scans/second
- <1cm range resolution
- Max range up to ~~50-80~~ m 300m
- Problems only with mirrors, glass, and matte black.



< 1000 €

~ 6000 €

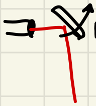
~ 40.000 €

first one on Google cars

> 80.000 €

Nowday for 500 € !

Lidar → you can have multiple laser but
because of cost I can Rotate laser
OR Rotate the light
by a mirror



You can moove
fast the light
LOTS measurements
Very accurate! ROBUST (lot energy)

all OUTDOOR vehicles work with LIDAR Reliable / accurate

not so costly sensors now, firstly HIGH COST!

- you can have multiple plane of mirrors scanned by device

→ 128 planes: 2 M points / sec!

Now
- small cheap sensors!
- more expensive industrial

LIDAR is expensive respect the others!

↑ data coming from Lidars is a lot used
for mapping / Localization

(!) dust / snow.. reflect the light! bad working
due to meteo

(RADAR) based on basic principle...
nice property → you can use doppler effect
to measure also velocity (speed) of obj
(used in cars)

↓ BUT noisy signal and expensive
NO used for ROBOTS → research on
Automomous drive
because see through rain, snow, ecc..

Position sensors (outdoor)

indor / outdor

GNSS

use GPS, GLONASS satellite for positioning

Outdoor position can be measure by a "Global Navigation Satellite System"

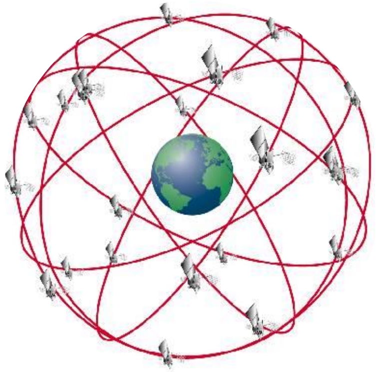
- Several constellations exist (GPS, GLONASS, Beidou, Galileo, …)

CHINESE European

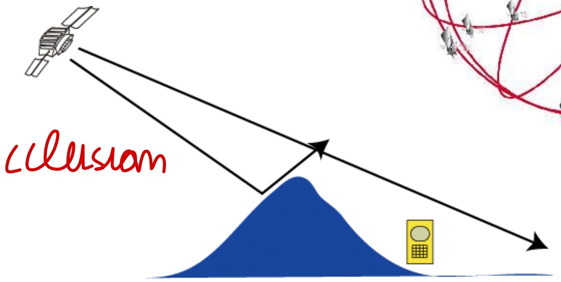
(american satellite) (american implementation of GPS)

Global Positioning System (GPS)

- 24 satellites orbit the Earth twice a day
- They synchronously emit location and time periodically position
- Receiver compares transmitted and arrival times
- At least 4 sensors must be perceived
- Accuracy is about 2.5m@2Hz (20 cm DGPS)

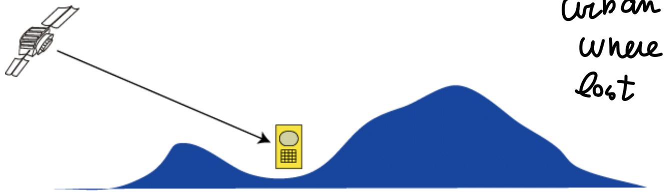


occlusion



No line of sight reception

"urban caniom..." where signal lost



Line of sight reception

Several issues

occlusion

- Do not work indoor, underwater, or in urban canyon
- Need line of sight reception
- Suffer multiple paths and reflections → like so many probleem... longer distance due to multiple path reflection or LOST SIGNAL

Know your position knowing the position of satellite ... the receiver from satellite distance can tell where is the receiver!

In Normal conditions → good signal !

with high accuracy

in Real time Kinematics ⇒ 2 cm error

sensor

HIGH COST!

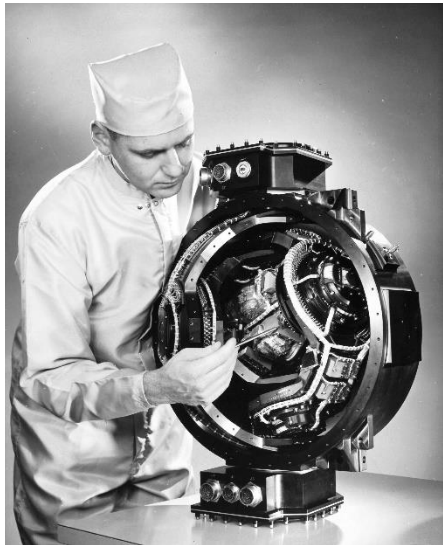
good precision

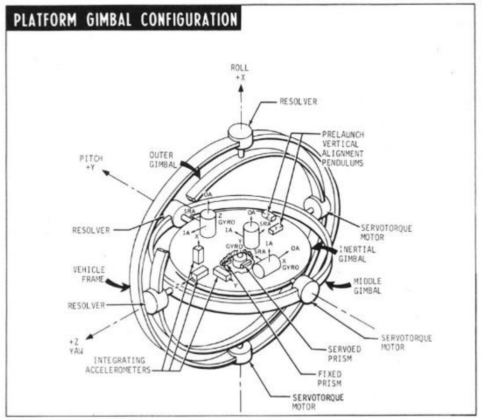
Inertial sensor

used for ROBOTS measuring forces, acceleration, inertia

- Gyroscopes
  - Angular velocities
- Accelerometers
  - Linear accelerations
  - Gravitational vector
- Magnetometers/compass
  - Earth magnetic field vector

ST-124 Inertial Guidance Platform used in the Saturn V, 1960s for SATURN Nasa mission





nowdays cheap sensor

An Inertial Measurement Unit (IMU) fuses gyroscopes, accelerometers and magnetometers to provide full 6DoF pose estimate

Intertial measurements integration (e.g., to compute position) cumulate errors and drifts significantly over time, especially with cheap MEMS technology …

Noise/accuracy very different!

lot used for DRONES thanks to control of
altitude, rotation
@ LOWER COST than expensive Giroscope

Tactile Sensors → for safety! when hitting it stops! thanks to "switches"

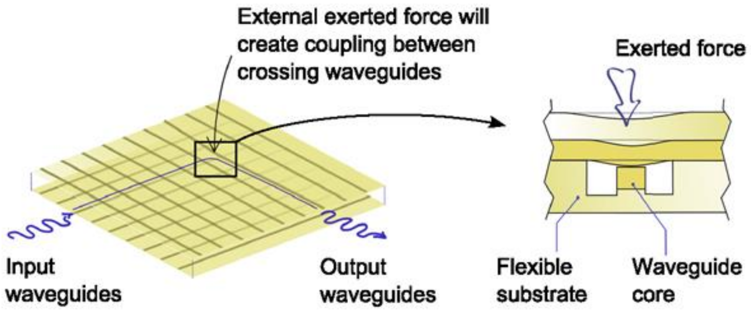
artificial skins
or CAPACITORS (like on smart phone)

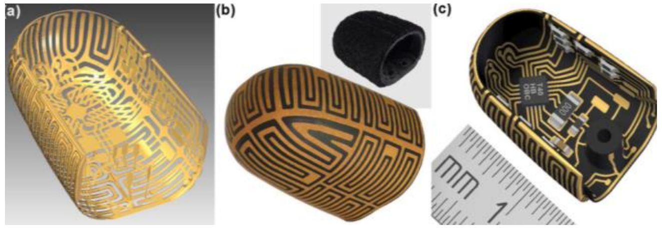
These sensors are used for manipulation purposes

measuring contact, or array of sensors to know where touching

Two main categories

- Binary
  - realized by switches
  - placed on the fingers of a manipulator
  - may be arranged in arrays (bumpers)
  - on the external side to avoid obstacles
- Analogical (real valued)
  - soft devices that produce a signal proportional to the local force
  - a spring coupled with a shaft
  - soft conductive material that change resistance with compression
  - measure also movements tangential to the sensor surface

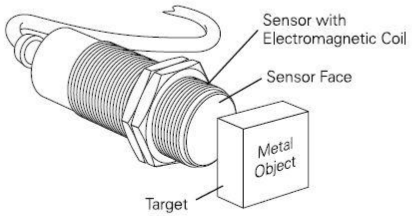


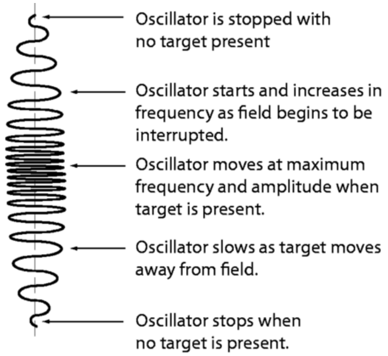


Proximity Perception (short range distance)

Measure the presence of objects within a specified distance range

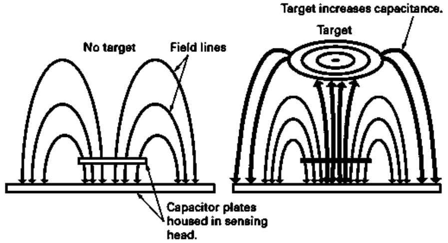
- Used to grasp objects and avoid obstacles

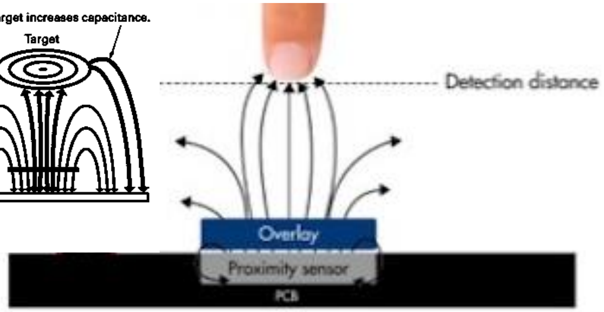


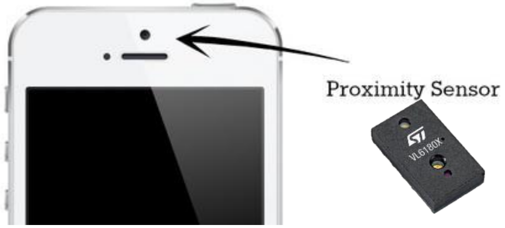


Several technologies:

- Ultrasonic (low cost)
- Inductive (ferromagnetic materials at distance <mm)
- Hall effect (ferromagnetic materials, small, robust, & cheap)
- Capacitive (any object, binary output, high accuracy when calibrated for a particular object)
- Optical (infrared light, binary or real valued)







Distance sensor

Mapping sensors → see obstacles, .. computer vision

image analysis